

TEST AND EVALUATION REPORT

Fake News Detection Using Social Media Text

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1. Introduction

This Test and Evaluation Report presents the evaluation process and observations of the Fake News Detection System. The purpose of this report is to analyze the system's performance based on defined test cases and scenarios. The evaluation focuses on classification behavior, system reliability, and response to different types of input data.

2. Evaluation Approach

The system was evaluated using manual testing based on predefined test cases and test scenarios. Different types of social media text inputs, including real news, fake news, and invalid inputs, were provided through the Streamlit interface. The system's predictions were observed and compared with expected outcomes to assess correctness and stability

3. Performance Metrics

- **Accuracy:**
The system demonstrated satisfactory accuracy in distinguishing between real and fake news for commonly observed social media text samples.
- **Precision:**
The model showed consistent precision in identifying fake news content with minimal false classifications.
- **Recall:**
The system was able to detect a majority of fake news samples provided during testing, indicating effective recall performance.

4. Test Results Summary

All defined test cases were executed successfully. The system correctly classified real and fake news inputs and handled invalid or empty input cases appropriately. No critical defects were identified during testing, and the application maintained stable behavior throughout the evaluation process.

5. Observations

- The system responded quickly to user inputs
- Classification results were displayed clearly on the user interface
- Error handling worked effectively for empty inputs
- The application maintained consistent behavior across multiple test runs

6. Limitations

- The model may misclassify ambiguous or sarcastic content
- Performance may vary with extremely long text inputs
- Accuracy depends on the quality and diversity of training data

7. Conclusion

The evaluation results indicate that the Fake News Detection System performs reliably under expected usage conditions. The system meets its functional requirements and demonstrates stable classification behavior. Overall, the application is suitable for deployment in controlled environments and further enhancement.